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Functional Data Analysis for Prediction of Employee Presence

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ABSTRACT The problem of space occupancy and the prediction of occupancy schedules has rarely been addressed, due to the highly stochastic behavior of occupants and the insufficient quality of data sets. The COVID-19 pandemic has challenged traditional models based on the physical presence of employees in the workplace. This crisis has accentuated the evolution of new trends, such as teleworking, and highlighted the inherent diversity of work environments, as well as the multitude of factors that influence employee deplacements. The ability to anticipate when people will come is essential for effective management of these spaces, particularly in terms of resource planning, safety, and comfort. As a result, predicting when employees will get to work is a complex challenge. Accurate presence prediction tools can optimize resource utilization and support strategic decisions in corporate real estate. This paper proposes a robust approach combining functional data analysis with the integration of exogenous variables adapted to time series (RobFTS). The purpose of this approach is to anticipate the future presence of employees in their workplace. The model is developed using lags of functional time series, as well as external variables such as weather data (temperature, precipitation) and meeting rooms reservations. In addition to standard error metrics, we apply a comprehensive model selection process using AIC, BIC, AICc, HQIC and other information criteria adapted for non-parametric models. This method is then applied to a real database, gathering daily observations of employee presence at a corporate site in Paris. Finally, we contrast this method with other models widely used in time series analysis, Seasonal Autoregressive Integrated Moving Average with Exogenous Regressors (SARIMAX), Long Short Term Memory (LSTM), Adaboost, and Deep Learning for functional data analysis (DFMM). It is found that the mean absolute percentage error (MAPE) is about **4.99%** for RobFTS, compared to **17.30%**, **12.92%**, **11%** and **6.50%** for SARIMAX, LSTM, Adaboost and DFMM, respectively. The data on which we applied the different models is made available on **GitHub**[a], along with the various scripts of the models.

[a]https://github.com/goujilinouhaila-coder/Prediction-of-Employee-Presence-in-Smart-Workplaces

INDEX TERMS New Ways of Working (NWoW), Functional Data, Occupancy Prediction, Exogenous Variables, Functional Time Series, Deep Learning.

I. INTRODUCTION

Today, the workplace is caught between the demands of new ways of working and the opportunities for enhanced comfort on one side, and the costs associated with the work environment during a challenging economic period for businesses on the other. Smart workplaces are transforming the future of work by offering a host of essential benefits and features. These cutting-edge work environments have substantial influence on workplace dynamics, resulting in heightened levels of productivity and efficiency. The revolution in Information and Communication Technologies (ICT) has driven the evolution of workplaces over the past few decades. The concept of smart offices has gained in popularity, mainly referring to sensor-equipped workplaces connected to the internet and mobile devices [1], [2], [3]. Many studies in the literature have focused primarily on user behavior through the collection of sensor data, with the aim of developing technology and improving the quality of smart workplaces. Additionally, occupancy data has been gathered with a focus on preferences to enhance workplace efficiency in smart workplaces, as discussed in [4]. In addition, sensor data has been used to forecast user expectations, particularly in providing thermal comfort within

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smart offices, as demonstrated by [5].

In the wake of the COVID-19 pandemic, the concept of New Ways of Working (NWoW) has become a focal point for organizations. These modes of work, enabled by advances in ICT, offer workers greater flexibility in managing their space and time [6], [7], [8]. The academic literature on this topic has experienced significant growth in recent years, particularly in the field of management sciences, aligning with organizational changes and the integration of new tools designed to augment work flexibility [9], [10]. Consequently, numerous recent studies have aimed to evaluate the potential advantages of NWoW for both employees and employers [11], [12]. Workspaces in particular, or working environments in general, are impacted by these new forms of hybrid working. Examples include flexible spaces such as flex office spaces or Activity-Based Workplaces (ABW). The ABW concept offers employees a variety of spaces, including different layouts, sizes, technological equipment, and different atmospheres. For example, it includes classic workstations, phone booths, quiet zones, relaxing areas, project, and activities spaces. People have the freedom to choose the environment that best suits their job, state of mind, and personal preferences. In an ABW, all spaces are theoretically accessible to everyone, regardless of hierarchical status or function within the organization. This approach often results in underutilized workstations as employees increasingly work remotely. And also with the success of remote working (Work From Home, WFH) during the pandemic, a decrease in demand for office space has been observed [13].

From a financial perspective, the costs associated with office space generally represent between 10 and 20% of the total cost of work. This expense can seem particularly high when occupancy rate are low, sometimes less than 30% since the development of post crisis telecommuting. Cost reduction can be achieved in a number of ways, such as optimizing location to take advantage of more advantageous rental or construction costs, which may involve frequent moves. Another approach is to reduce the surface area assigned to each professional situation (activity) [14]. Organizations can reduce real estate costs by optimizing space utilization, identifying opportunities to reduce leased space, or negotiating more favorable lease terms. During 2019-2020, significant variations were seen in the composition of total workspace costs in Paris and the Île-de-France region. On average, costs per square meter amounted to around 732 euros, per workstation 13,658 euros, and per employee 12,726 euros. These numbers figures represent decreases of around 1% per square meter and 2% per workstation compared to the period 2017-2022. However, between 2021 and 2022, workspace expenditure recorded an increase of 8% per square meter, 9% per shift and 10% per employee [15]. The pressure of limited square meters and rising real estate costs are driving organizations to optimize and maximize space efficiency. Regardless of the working method adopted, measuring occupancy remains essential.

All these considerations lead us to consider the question of workspaces, and we seek to identify the issues involved in modeling office worker behavior through the real estate dimension. Since real-estate costs are significant for organizations, but the emergence of more flexible work arrangements makes the occupation of workplaces more uncertain, organizations need to equip themselves with suitable tools. These tools should primarily focus on the quantitative aspect, facilitating the adjustment of real estate resources based on the number of individuals utilizing workspaces. Achieving this requires the ability to gather pertinent information and data.

To address these challenges, this study presents a new approach based on Robust Functional Time Series Analysis (RobFTS), enriched with exogenous variables such as weather and meeting room bookings, to improve the accuracy of employee presence prediction. This innovative methodology not only improves predictive performance, but also provides actionable information for optimizing workplace management, as validated through real-world experiments.

While classical function-on-function regression models were originally developed for continuous function prediction, robust functional models with exogenous variables have been predominantly applied to classification tasks in practical applications. In this study, we adapt these robust functional approaches to a time series prediction setting, integrating exogenous information to forecast employee presence over time.

The main contributions of this work are summarized as follows:

1) We publicly release a real-world dataset of employee presence, including weather data and meeting room usage, to encourage reproducibility and facilitate future research in workplace analytics.
2) We adapt the model proposed in [16] to the context of functional time series, and refer to this adapted version as **RobFTS**, which we apply to the prediction of employee presence using functional data with exogenous variables.
3) We extend a functional deep learning model, initially designed for classification, to the context of time series forecasting, introducing necessary modifications to handle continuous target variables and temporal dependencies.

The remainder of the paper is structured as follows. Section II provides a review of the literature on occupancy and functional data analysis. Section III presents the methodology, outlining the different models applied. It begins with an introduction to Functional Time Series (FTS) models, focusing on the robust function-on-function approach with RobFTS, followed by a discussion of Deep Learning for Functional Data Analysis and the functional deep learning model. Section IV covers the application to the case study, including a description of the data, data processing steps, and the system proposed in our study. The results are presented in Subsections IV-D, where the SARIMAX, LSTM, AdaBoost and the functional models are evaluated. Subsection IV-F details the information criteria used for model comparison across

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the different configurations, along with their corresponding results. Section V presents a comparison with public data on the occupancy rate of parking lots. In Section VI, we discuss the limitations of our approach, including computational cost. This section also examines the implications of these results concerning real estate industry constraints and the optimization of workspaces as an economic challenge. Finally, Section VII presents the findings and concludes the paper.

II. LITERATURE REVIEW

A. LITERATURE REVIEW METHODOLOGY

A structured literature review was conducted to ensure a comprehensive and systematic analysis of prior research on workspace occupancy prediction using machine learning and functional time series analysis. This section details the methodology used to select and compare relevant articles.

1) Databases and Search Strategy

To retrieve relevant studies, we queried multiple academic databases, including IEEE Xplore, ACM Digital Library, Scopus, Web of Science, and Google Scholar. The search strategy was defined using a combination of keywords related to **occupancy prediction, machine learning, functional data analysis, and time series forecasting**. The following Boolean query was applied:

> (("occupancy prediction" OR "workspace analytics") AND ("machine learning" OR "deep learning") AND ("functional data analysis" OR "time series forecasting"))

The search was conducted for peer-reviewed journal articles and conference papers published between 2015 and 2024. Regarding the inclusion and exclusion criteria, only studies published in reputable journals or conferences, focusing on workspace occupancy prediction using machine learning and validated with real-world data, were included. The selection excluded purely theoretical works, non-reproducible studies, and publications in languages other than English. Additionally, studies addressing topics unrelated to occupancy prediction, such as energy consumption forecasting, or those focusing on classification problems rather than regression, were also excluded.

2) Article Selection Process

The article selection followed the **PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines**. Initially, a total of 120 articles were identified in different databases. After removing duplicate entries (30 articles), 90 articles remained for selection based on their title and abstract. 50 articles were excluded because they were purely theoretical, non-reproducible, published in languages other than English, or classification-only problems rather than regression. 25 additional studies were excluded for not specifically addressing occupancy prediction in corporate buildings, resulting in a final total of 15 studies included in our review of the review of the literature. The PRISMA flow diagram illustrating this process is shown in Fig.1.

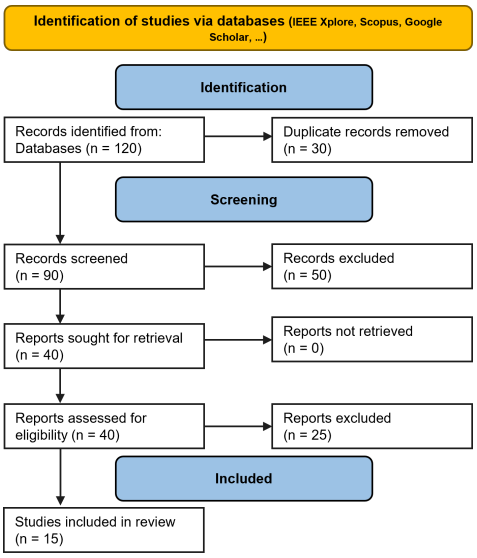


FIGURE 1. PRISMA flow diagram illustrating the article selection process [17].

B. OCCUPANCY ANALYSIS & PREDICTION

As explained above, the effective management of workspace occupancy is an essential subject for study. It has a significant impact on the financial costs and sustainability of working environments. In-depth occupancy analyzes can help reduce these costs, both in the design phase and during building use. Machine learning-based models have proven to be fast and accurate tools for predicting occupancy. However, during the COVID-19 pandemic, office occupancy changed radically.

In [18], the authors provide a comprehensive review of the key challenges in occupancy prediction, highlighting the necessity of hybrid methodologies that integrate sensor-based data, machine learning techniques, and statistical modeling to improve predictive accuracy. Their study identifies functional time series models and deep learning architectures as particularly effective in capturing temporal variations in occupancy patterns. These approaches demonstrate considerable potential to improve the precision of occupancy forecasts.

The study in [19], analyzes the occupancy densities of two buildings during different pandemic periods. For Building A, the occupancy density never exceeded 0.049 people / m2 and for 92% of the time it did not exceed 0.016 people / m2. In Building B, occupancy peaks at 0.113 people/m2, but the range from 0.066 to 0.0113 people/m2 is only recorded 15% of the time. The dominant occupancy density for both buildings during pandemic periods is 0.016 people/m2. The study shows that the actual office occupancy was well below the standards and methodologies, even when quarantines were canceled. The average peak occupancy throughout the measured period was 12-20% for Building A and 2-23% for Building B.

In [20], the authors propose an approach called BLEMAT (Bluetooth Low Energy Microlocation Asset Tracking) for indoor positioning, incorporating new techniques for data trans-

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formation, fusion and analysis, along with machine learning methods, to map indoor location data to building occupancy patterns. The authors model the datasets from previously validated indoor positioning results as time series to predict occupancy patterns. They then train the Long Short-Term Memory (LSTM) neural networks to make these occupancy forecasts. The research is aimed at providing valuable insights for efficient resource management and cost-saving strategies in the observed residential building.

In [21], the authors have proposed an OccupancySense model, which aims to improve the accuracy of occupancy detection and prediction. It combines indoor air quality (IAQ) data based on Internet of Things (IoT) with static and dynamic contextual data for greater prediction accuracy. The model uses CatBoost, a machine learning algorithm that is used for classification and regression tasks. The algorithm uses the concept of ordered reinforcement, which reduces information loss and overfitting while managing categorical features. To achieve this, it uses target statistics (TS) and a ranking principle. The algorithm constructs decision trees as basic predictors and uses balanced trees to speed up execution and testing time. Overall, CatBoost is a powerful algorithm that combines the advantages of gradient amplification with efficient management of categorical features, making it suitable for a variety of machine learning tasks. The OccupancySense model is non-intrusive, but has high predictive power. It can accurately detect occupancy levels, predict staffing levels, and estimate the room occupancy density with an accuracy of 99.85%, 93.2% and 95.6% respectively, as confirmed by a real-life case study with 91 days of continuous indoor data. In the following sections, we will explore functional data analysis techniques, as their advantages allow for better capturing of temporal dynamics and modeling the continuous variations in occupancy rates over time.

Beyond machine learning techniques, robust statistical methods have been increasingly explored for occupancy prediction in time series data. Functional data analysis (FDA) has emerged as a powerful tool to model continuous occupancy variations over time. Regression models such as functional principal component regression (FPCR) [22] and functional autoregressive models (FAR) [23] have been proposed to handle high-dimensional functional predictors.

Given these findings, we adopt a functional time series approach to capture occupancy variations while incorporating exogenous variables such as weather conditions, public transportation strikes, and meeting room bookings. This approach allows for better generalization and interpretability, making it a robust alternative to purely data-driven methods. In the following sections, we introduce different approaches of Functional Data Analysis.

C. FUNCTIONAL DATA ANALYSIS

Functional Data Analysis (FDA), extensively explained by [22], includes a set of statistical techniques designed to handle the analysis of curves and surfaces formed by smoothing discrete data series. Introducing this complexity proves advantageous, as curves offer a more intuitive approach to modeling problems with a regular phenomenon [24]. FDA is a statistical discipline that takes advantage of the functional nature of data [22], [24]. In the context of venue prediction in workplaces and buildings, where data can be collected from sensors or other automated sources, the accumulation of large data sets makes this an attractive field for functional approximation and representation.

In our context, our aim is to predict a series of functional data observed over a period of time. These functions are recorded (with errors) at various time points ranging from $t = 1$ to $t = T$, where T represents the observation period, which can be a day, a week, a month, or multiple time intervals. Our goal is to forecast these functions for future time points that extend from $t = T + 1$ to $t = T + h$, where h is the time period we want to predict, and we will use functional time series for this purpose.

Several studies on functional time series exist in the literature. In the article by [16], the authors introduced a functional autoregressive model (FARX(1)) by incorporating exogenous hydrometeorological variables to predict river flow. This FAR model integrates lagged variables as well as essential exogenous climate variables, such as precipitation, temperature, and evaporation, to capture the inherent correlations and dynamics in river flow time series. The FARX(1) model extends the FAR(1) model by including lagged data and additional exogenous variables, improving the accuracy of future river flow forecasts. The Mean Absolute Percentage Error (MAPE) was calculated to evaluate the prediction performance of the models. In the article [23], the authors introduced a functional approach to ozone forecasting. This study incorporates exogenous variables, notably wind speed and temperature, into Hilbertian autoregressive models. The performance of these models was evaluated based on 24-hour-ahead forecasts of hourly ozone concentrations, with a particular focus on predicting maximum daily ozone concentrations. The results show that Hilbertian autoregressive models, enriched with exogenous variables, outperform the kernel functional model in terms of predictive power for ozone forecasting, with a Mean Squared Error (MSE) of **239.4** and a Root Mean Squared Error (RMSE) of **15.47**. Furthermore, the article [25] by Hyndman and Shang proposes methods for forecasting functional time series (FTS) by applying two key approaches: Weighted Functional Principal Component Regression (FPC) and Weighted Functional Partial Least Squares Regression (FPLS). Functional time series data refers to a sequence of curves observed over time, such as mortality rates or fertility rates. The goal of the article is to forecast future curves based on past data, while accounting for the time series nature of the data. The paper concludes that using weighted versions of both FPC and FPLS improves forecast accuracy, especially when dealing with long time series data. The FPLS method is particularly advantageous due to its focus on forecasting and reduced computational cost. Both methods are implemented using the tools available in the *ftsa* R package developed by the authors, as detailed in Subsection III-A3, which we

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applied to our case study.

D. ROBUST FUNCTIONAL METHODS

Classical functional methods, while powerful for modeling data evolving in a functional space, are often sensitive to outliers and the noise inherent in real observations. To meet these challenges, several studies have proposed robust variants adapted to functional data. These include robust regression techniques [26], robust functional principal component analysis (FPCA) [27], [28] and robust functional clustering [29]. The aim of these methods is to guarantee a better estimate of the underlying structures without being strongly influenced by extreme observations. Robust functional principal component analysis (robust FPCA) has thus been developed to improve data representation in noisy contexts, enabling better dimension reduction and preservation of essential trends in observed functions [30], [31]. In addition, robust clustering and classification methods for functional data have emerged, improving the reliability of segmentation of complex data in the presence of noise [29]. Finally, some approaches have focused on functional anomaly detection, where robust models can identify outliers without compromising the overall quality of the analyses performed [32].

Although many functional data methods are effective, some applications require robustness against outliers and noise. Several approaches in the literature have addressed this issue by developing **robust functional regression models and functional autoregressive models.**

In the article [26], the authors presented a functional linear regression model using functional principal component regression estimators, both robust and least-squares-based. The performance of this model depends on the number of principal components used. Through a simulation study, the authors successfully captured regression behavior in the presence of outliers, accurately identifying these atypical observations.

To summarize the different approaches of these articles, in [23], the authors introduced the FARX(1) model, incorporating both lagged and exogenous variables (e.g., precipitation, temperature) for river flow forecasting. This model improved accuracy compared to simpler models by capturing the effects of these external factors. Similarly, in [16] a functional approach is applied to the prediction of ozone, using exogenous inputs like wind speed and temperature to outperform kernel-based models. In conclusion, while each model integrates exogenous variables to improve predictive power in functional data contexts, our approach focuses on adapting these methods to time series by leveraging lagged values of the response variable alongside exogenous predictors. This combination better captures dynamic interactions over time.

III. METHODOLOGY

In this section, we will describe the methods evaluated in this paper: Functional Time Series models and Deep Learning for Functional Data Analysis. We propose an approach based on the functional linear regression (FLR) model, which is adapted to time series to predict the presence of employees in buildings and workplaces. In our approach, we apply the FLR model to time series, which typically models the variable y exclusively based on explanatory variables x. However, our proposed approach aims to establish a relationship with the variable y (e.g. the number of people present in the building) not only as a function of the explanatory variables x, but also with the lagged values of y.

A. FUNCTIONAL TIME SERIES (FTS) MODELS

1) Functional linear regression (FLR)

In this section, we give a brief summary of the FDA tools that we will later apply in our model, namely the Functional Linear Regression (FLR) model. In the following sections, we will use the vector space $L^2(I)$ which is the Hilbert space of square integrable functions on the compact interval I with the inner product $\langle f,\ g\rangle = \int_I f(t)g(t)dt$ for functions $f, g \in L^2(I)$.

Let $X(t)$, $Y(t)$ be stochastic processes defined on I, with mean functions $\mu^X(t)$ and $\mu^Y(t)$. Define $x(t) = [x_1(t), ..., x_n(t)]$ and $y(t) = [y_1(t), ..., y_n(t)]$ as n independent and identically distributed realizations of $X(t)$ and $Y(t)$, respectively.

The selected and applied model in our case study is the robust function-on-function regression model [26], which we will adapt to time series. This model employs function-on-function regression models. In the traditional Functional Linear Regression (FLR) model introduced by Ramsay and Dalzell in 1991 [24], we model the relationship between predictor $x_i(t)$ and response $y_i(t)$ as in (1):

$$y_i(t) = \alpha(t) + \int_I x_i(s)\beta(s,t)\mathrm{d}s + \varepsilon_i(t). \tag{1}$$

Here, $\alpha(t)$ represents the intercept function, $\beta(s,t)$ is the parameter function to be estimated, and $\varepsilon_i(t)$ denotes the error process. When considering a fixed value of t, $\beta(s,t)$ can be interpreted as the relative weight placed on $x_i(s)$ to predict $y_i(t)$. To build data-driven basis functions for $x_i(s)$ and $y_i(t)$, [26] used Functional Principal Component Analysis (FPCA). FPCA is a method used to identify the dominant modes of variance of functional data. These dominant modes of variance are called Functional Principal Components (FPCs). FPCA also serves as a tool for reducing dimensionality, enabling effective approximation of a set of observed curves through a concise representation using a small set of Functional Principal Components (FPCs) in a linear combination.

As stated in [33], we will adopt the assumption that the mean functions $\mu^X(t) = 0$ and $\mu^Y(t) = 0$, implying $\alpha(t) = 0$. This assumption is reasonable in practical scenarios, where the mean functions $\mu^X(t)$ and $\mu^Y(t)$ can be efficiently computed for dense data. Subsequently, the data can be preprocessed by subtracting $\mu^X(t)$ and $\mu^Y(t)$ from the observed curves.

To illustrate this approach, let's consider our case example:

- We want to predict the number of employees present in an office building on a given day or a period.

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- The attendance $y_i(t)$ is the prediction for the upcoming period, for example for the term n+1.
- $x_i(s)$ is the lagged variable of order 1.

2) Robust function-on-function approach to time series with RobFTS

Based on the FLR model detailed in the previous section and the robust function-on-function approach, we propose an approach that adapts this model to time series (RobFTS). We can apply the RobFTS model to predict the presence of workers in the tertiary building. Let $(Y_n)_{n=1,\dots,T}$ with for each n, $Y_n \in L^2(I)$, the functional time series of the number of workers by week. Let define $\left(X_n^k\right)_{n=1,\dots,T}$ with $k = 1, \dots, K$ the K-th exogenous FTS with $X_n^k \in L^2(I)$. The model is given in (2) as:

$$Y_{n+1}(t) = \sum_{l=1}^{L} \int_I \beta_l(s,t) Y_{n+1-l}(s)\mathrm{d}s + \sum_{k=1}^{K} \int_I \gamma_k(s,t) X_n^k(s)\mathrm{d}s + \varepsilon_n(t), \tag{2}$$

where L is the number of lagged variables used, β_l for $l = 1, \dots, L$ and γ_k for $k = 1, \dots, K$ are the functional parameters to be estimated.

In the context of Equation (2), let us consider the case where there is only one lag, $L = 1$, and three exogenous variables, $K = 3$, the model simplifies to:

$$Y_{n+1}(t) = \int_I \beta_1(s,t) Y_n(s)\mathrm{d}s + \sum_{k=1}^{3} \int_I \gamma_k(s,t) X_n^k(s)\mathrm{d}s + \varepsilon_n(t). \tag{3}$$

In this example:

- We want to predict the number of employees present in an office building on a given day or an upcoming period Y_{n+1} (e.g., employee attendance for the next week).
- The attendance $Y_n(s)$ is influenced by different factors (K=3 in this case):
  Weather conditions: Employees might avoid coming to the office on rainy days.
  Transportation strikes: If a subway strike occurs, attendance might be significantly lower.
  Meeting room reservations: A high number of meeting room bookings may indicate a greater presence in the office.
- Each of these factors is represented as an explanatory functional variable $X_n^k(s)$, which the model integrates over time to estimate future values of Y_n.
- In this way, by incorporating both the information of the previous period (Y_n) and various contextual variables (X_n^k), we seek to capture the dynamic and exogenous effects that influence the future outcome.

To ensure robustness in parameter estimation, we compare multiple robust estimators in the RobFTS framework. These include:

1) **MCD (Minimum Covariance Determinant)**, as introduced in [34], which minimizes the determinant of the covariance matrix for robust regression.
2) **MM Estimator** proposed in [35], which provides high breakdown robustness while maintaining efficiency.
3) **tau Estimator** from [36], which balances robustness and efficiency, offering resistance to outliers while maintaining good statistical properties.
4) **S Estimator** developed in [37], which is designed for high breakdown robustness and reliable resistance to leverage points.

Once the functional parameters β_l and γ_k have been estimated from historical data (yielding $\widehat{\beta}_l$ and $\widehat{\gamma}_k$), the prediction for $Y_{n+1}(t)$ is obtained by substituting these estimated coefficients and the previously observed values of $Y_{n+1-l}(s)$ and $X_n^k(s)$ into Equation (3). Specifically, the forecast can be written as:

$$\widehat{Y}_{n+1}(t) = \int_I \widehat{\beta}_1(s,t)\, Y_n(s)\, \mathrm{d}s + \sum_{k=1}^{3} \int_I \widehat{\gamma}_k(s,t)\, X_n^k(s)\, \mathrm{d}s. \tag{4}$$

Here, $\widehat{Y}_{n+1}(t)$ represents the prediction of the model of $Y_{n+1}(t)$ given the information available at time n. This framework allows us to integrate lagged values of the target variable, as well as multiple exogenous functional predictors, into one unified forecasting approach.

3) Forecast functional time series with *ftsa*

We compared various method of the *ftsa* package [25], for modeling and forecasting functional time series. This model utilizes optimal orthonormal principal components obtained from a principal component decomposition. The method used is "rapca", which means that the robust principal component analysis algorithm of [38] is used.

The coefficients of the fitted model are forecasted using either an ARIMA model (method="arima"), an AR model (method="ar"), an exponential smoothing method (method= "ets"), a linear exponential smoothing method allowing missing values (method="ets.na"), or a random walk with drift model (method="rwdrift"). The forecast coefficients are then multiplied by the principal components to obtain a forecast curve.

The limitation of this method is that we do not include exogenous variables such as temperature, rainfall, etc., unlike other methods. We apply the method only using employee presence data.

B. DEEP LEARNING FOR FUNCTIONAL DATA ANALYSIS (DFMM)

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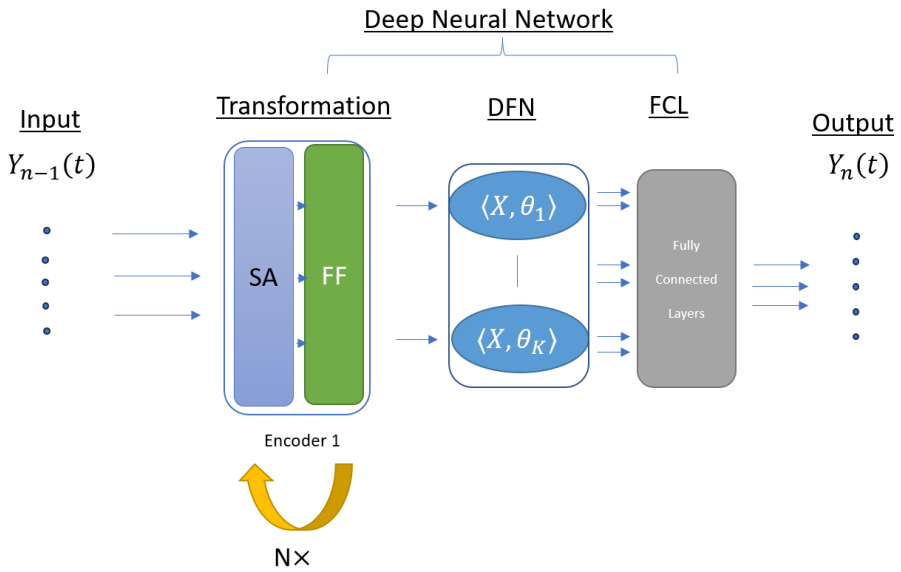


FIGURE 2. Proposed method. SA: Self-Attention, FF: Feed-Forward network, DFN: Deep Functional Network, FCL: Fully Connected Layers.

1) Functional multiple index models

We adapt the recent article of [39] designed for classification task to our case. Let us introduce the model:

$$(Z_n^p)_{p=1,\dots,L+K} = f(Y_n, \dots, Y_{n+1-L}, X_n^1, \dots, X_n^K), \quad (5)$$

$$Y_{n+1} = g\left(\langle Z_n^1, \theta_1 \rangle, \dots, \langle Z_n^p, \theta_{L+K} \rangle\right) + \varepsilon_n, \quad (6)$$

where $f : (L^2[0,I])^{L+K} \to (L^2[0,I])^{L+K}$ is a transformation of the multivariate functional data, θ_j are the indexes (functions of $L^2[0,I]$), $g : \mathbb{R}^{p\times K} \to L^2[0,I]$ is the link function, $\langle .,. \rangle$ is the inner product of $L^2[0,I]$ functions and the power j of the variable Z_i is the j-th component function of Z_i.

The unknown functions are the transformation T, the link function g and the indexes $\theta_j, j = 1, \dots, L+K$. The transformation T is inferred by a transformer encoder architecture. Next, we apply an AdaFNN network to each functional variable. Thus, θ_j are estimated using a deep functional network (DFN) based on a concatenation of the AdaFNN module. We apply a fully connected layer to estimate the link function g. Finally, we regularize the output by a L^2 loss of the square of acceleration of the parameter to output the final Y_{n+1}.

2) Deep neural network of the model

The article of [39] is designed to classify emotions in speech based on Mel Frequency Cepstral Coefficients (MFCCs) that can be regarded as a multivariate functional object. Here, we adapt this network to the regression task, because the model was initially designed for a classification problem and is not directly applicable to regression problem. The transformation module is made of a stack of N transformer encoders with self-attention and a feed-forward network. The second module is the neural network of the paper [40] which is generalized to adapt to the multivariate context. In few words, the paper of [40] proposes to integrate the data by an adaptive function by a numerical scheme. So, the outputs of this second module are simply the L^2 product between each component of the transformed variables and an adaptive function. The third module of our network is a classical fully connected layer. The proposed method is represented globally in Fig. 2.

The four modules of the network architecture are:

1) The transformation module is made of N transformer encoders.
2) The Deep Functional Network that outputs the scores of multivariate functional variable.
3) Fully Connected Layers to reduce the dimension.
4) A regularization module with a L^2 loss on the acceleration of parameter to produce the output.

The last module is not represented in Fig. 2 as it is more a loss term added to the general loss than a proper network. There is three loss, the L^1 and L^2 loss of all the AdaFNN sub-modules as in [40] and the regularization loss (ie. L^2 of the acceleration of the parameter to output the final Y_{n+1}).

IV. CASE STUDY

A. DATASET DESCRIPTION

We applied our models to real data collected from a software publisher offering products for trading, risk management, and transaction processing for participants in financial markets. This data was gathered at their Paris site, a 9-story office building with employees distributed across 4 divisions (teams). The building is equipped with access control doors, where employees use their badges to enter the premises. We collected this badge data over a one-year period, from September 2022 to September 2023.

In Fig. 3, we show an overview of the functional venue data in our real-life case study, segmented by day of the week, from

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1 (Monday) to 5 (Friday).

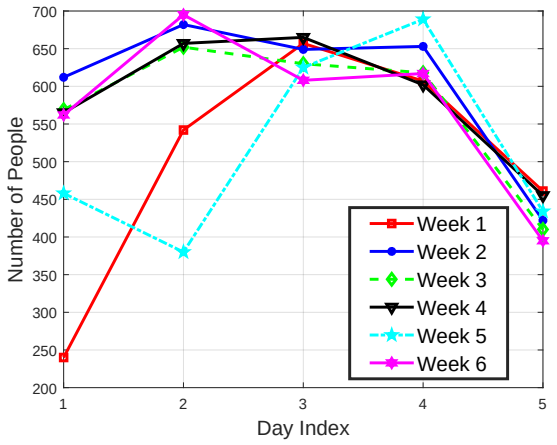


FIGURE 3. Functional time series plot of number of people by day of week. Each curve represents a different week, with colors distinguishing the weekly trends.

B. DATA PROCESSING

We processed badge data to calculate the unique number of employees accessing the site each day, excluding duplicates caused by repeated badge scans. In other words, we carefully examine the entry of each user to ensure that only unique and valid entries are taken into account. For example, if an employee badges in the morning and then goes out for a break, we do not consider this exit as a separate event. Our aim is to eliminate duplicate badges and to recover only unique and verified entries for each employee. Days with fewer than 100 people (8% of employees) are excluded from the dataset. This is considered to be a day when there was a problem on site, for reasons of data reliability and for stationary issues. This exclusion criterion was established based on business context constraints. In some cases, extremely low occupancy levels were observed due to external operational issues such as building maintenance, technical failures, or access restrictions. These events do not reflect regular office occupancy behavior and were therefore removed to avoid introducing outliers and noise into the predictive model.

For our data period, from September 1, 2022 to September 29, 2023, we process the data to obtain complete weeks during this period, excluding those where a day of the week is missing. For example, we ensure that each week covers Monday to Friday. From this processing, we obtain a database that contains 46 weeks.

We retrieved open-source weather data, including temperature and rainfall. In addition, strike and transport traffic data is available from a French government website.

In this case study, we retrieved meeting room reservation data from the office 365 calendar of the company, in order to identify for each day the number of reservations they had that day. The Table 1 provides an overview of the data frame that groups information on presence of people, weather, and meeting room reservations.

As part of our preprocessing, we excluded approximately 1% of the observations. These removed entries were not considered outliers, but rather corresponded to days with atypical conditions or incomplete activity, such as periods of exceptional building closure, which we had already justified earlier. This step was necessary to ensure the consistency of the model input with regular office activity patterns.

Given the nature of the data sources used in this study, no specific treatment for missing data or anomalies was required. The weather data and strike information were obtained from public APIs and governmental sources that provide validated and continuous datasets. The meeting room reservation and badge data were extracted directly from corporate systems designed for operational tracking, ensuring completeness and reliability. Consequently, the datasets used in our modeling process contained no missing observations, and no additional imputation or outlier correction was necessary.

1) Selection of Exogenous Variables

The selection of exogenous variables in our model was guided by data availability and relevance to occupancy dynamics. We leveraged the datasets that were accessible while ensuring that they provided meaningful contextual information for predicting employee presence. Specifically, we incorporated two main categories of exogenous variables:

- Contextualized workplace data: - Meeting room reservations: These serve as a key indicator of employee presence, as scheduled in-person meetings often correlate with higher workplace occupancy.
- External environmental factors: - Weather conditions: (e.g., heavy rain, snow) can impact commuting behavior and office attendance.

C. PROPOSED SYSTEM

Our proposed system starts with the collection of initial data, such as access badge data, meeting room reservation data, and weather data. These initial data, structured by dates, are transformed into a functional format. For instance, the data are restructured so that each row represents a week and each column represents a day of the week (from Monday to Friday). The restructured data are then divided into two sets: the training data, which are used to train the models, and the test data, which are used to evaluate the performance of these models. A loop is set up to evaluate the prediction error. In each iteration, one additional week is added to the training data, and the prediction error is calculated for the following week of the test data. This process is repeated until all weeks of test data have been evaluated. The average of the prediction errors obtained in each iteration is calculated to obtain an overall measure of the performance of the model. The proposed system is represented in Fig. 4.

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TABLE 1. Overview of the final dataframe that groups together presence, weather, meeting room reservations data and strikes: 1 if strike, 0 otherwise.

Date	N° of people	Temperature	Precipitation	Total reservations	Strikes
2022-09-05	240	20	3.1	186	0
2022-09-06	542	17	0.4	241	0
2022-09-07	657	17	0.5	212	0
2022-09-08	607	15	0.4	171	0
2022-09-09	461	15	0.4	140	0

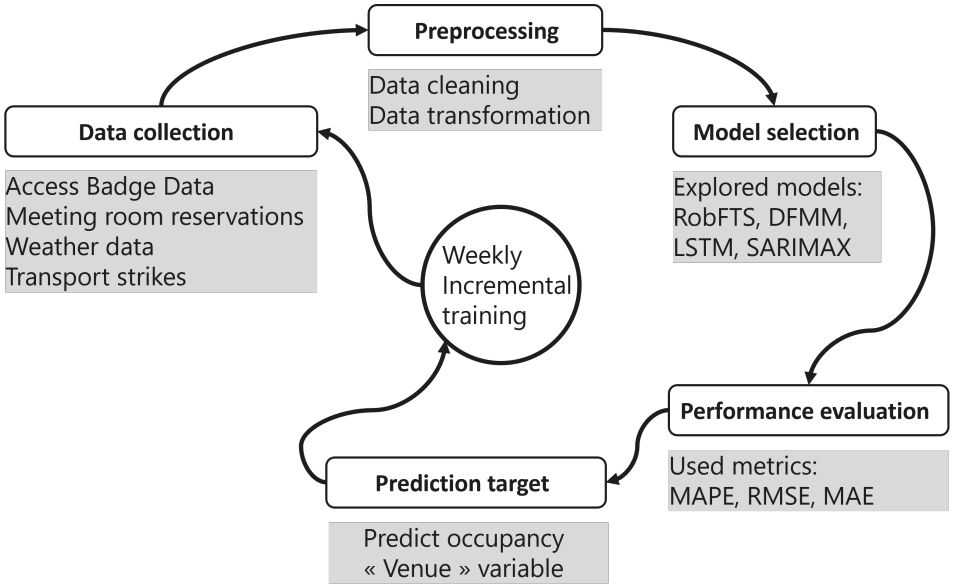


FIGURE 4. Proposed system for the prediction of occupancy.

D. RESULTS: PREDICTION OVER THE LAST 4 WEEKS

In this section, we present the results of the functional models described in Section III. We start by evaluating some classical approaches, including SARIMAX, as well as more recent models such as LSTM and AdaBoost, in order to assess their performance over the last four weeks of data.

For all models, we estimated the reliability of the metrics (MAPE, MAE and RMSE) obtained for each model, by repeating the entire modeling and prediction process a certain number of times (here 50). At each iteration, we collect one value of the metric, producing a sample of 50 estimates. The mean of this sample provides the overall estimate of the metric, and the 95% confidence interval (CI) is calculated according to the formula $\bar{x} \pm t_{\alpha/2,n-1} \times \frac{s}{\sqrt{n}}$, where $\bar{x}$ is the observed mean, s the standard deviation and n the sample size. We then present these intervals as "mean $\pm$ half-width of CI", to quantify the uncertainty around each performance measure.

1) Evaluation of the SARIMAX model

The model: We have implemented a SARIMAX model that extends the classic AutoRegressive Integrated Moving Average (ARIMA) approach by incorporating a seasonal component as well as exogenous variables likely to refine the forecast (in this case, for example, temperature, precipitation or the number of meeting room reservations). In our approach, each configuration of exogenous variables was evaluated according to the MAPE, MAE and RMSE metrics.

The SARIMAX model is an extension of the ARIMA model, which takes into account both the seasonal component of time series and exogenous variables (external factors that can influence the variable to be predicted). In addition to modeling series-internal autocorrelation (like ARIMA), SARIMAX manages seasonal effects and incorporates the potential impact of other explanatory variables, which can improve forecast accuracy [41].

Selection of SARIMAX Model Parameters:
To determine the optimal orders for the SARIMAX model, we conducted a grid search for the non-seasonal parameters (p, d, q) and the seasonal parameters (P, D, Q, s), where $s = 5$ reflects the weekly seasonality (assuming a 5-day working week). The values explored for each parameter were $p, d, q, P, D, Q \in \{0, 1\}$, yielding a total of $2^6 = 64$ seasonal

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and non-seasonal combinations. In this context:

- p is the order of the autoregressive (AR) component, representing the number of lagged observations used to predict future values,
- d is the degree of differencing required to make the time series stationary,
- q is the order of the moving average (MA) component, reflecting the number of lagged forecast errors included in the model.

Similarly, P, D, Q represent the seasonal counterparts of these parameters, capturing patterns that repeat every s periods (e.g., every week). We explored different values of p, d, q, P, D, Q. Each configuration was fitted to the training data, and the Akaike Information Criterion (AIC) was used as the selection criterion. The model minimizing the AIC was retained, as it provided the best trade-off between model complexity and goodness of fit.

Application & Results: To assess the robustness of the SARIMAX model over a four-week period, several combinations of exogenous variables were tested. The results, summarized in Table 2, show that integrating all the variables – Presence (Pr.), Temperature (T.) + Precipitation (P.) and Reservations (MR.) – gives a MAPE of around **17.3 ± 9.4**, a MAE of around **108.1 ± 59.1** and a RMSE close to **126.7 ± 65.4**. Conversely, reducing the set of variables (e.g. limiting to Pr., T. + P.) significantly degrades performance (MAPE to 35.2 ± 20), underlining the positive impact of the additional exogenous information.

TABLE 2. Table of SARIMAX model errors for each variable choice for prediction of four weeks. Pr.: for employee presence variable, T.: for temperature variable, P.: for precipitation and MR.: for meeting room reservations.

Choice of variables	MAPE	MAE	RMSE
Pr., T. + P. + MR.	**17.30 ± 9.40**	**108.14 ± 59.10**	**126.73 ± 65.44**
Pr., T. + P.	35.20 ± 20.00	216.02 ± 123.78	238.98 ± 124.53
Pr., T.	35.70 ± 21.20	222.68 ± 132.59	247.67 ± 136.09
Pr. + MR.	18.80 ± 10.60	116.93 ± 66.17	132.66 ± 69.68
Pr., T., MR.	18.00 ± 9.60	109.80 ± 59.16	129.04 ± 64.40

2) Evaluation of the LSTM model

The model: To enrich our comparison of forecasting models, we implemented an LSTM model [42]. This model is a type of recurrent neural network designed to handle sequences of data, particularly when long-term dependencies exist in the time series. In our study, the LSTM architecture deployed comprises two recurrent layers - each with a number of neurons adapted to the dimensionality of our data - followed by a final dense layer for predicting the target variable of presence.

Application & Results: We evaluated different scenarios by integrating the same combinations of exogenous variables. The Table 3 show the results obtained for the prediction over the last 4 weeks.

TABLE 3. Table of LSTM model errors for each variable choice for prediction of four weeks. Pr.: for employee presence variable, T.: for temperature variable, P.: for precipitation and MR.: for meeting room reservations.

Choice of variables	MAPE	MAE	RMSE
Pr., T. + P. + MR.	13.46 ± 0.46	85.10 ± 3.36	106.06 ± 4.05
Pr., T. + P.	19.48 ± 0.33	121.90 ± 2.83	139.27 ± 3.44
Pr., T.	20.27 ± 0.44	128.45 ± 3.50	147.89 ± 4.24
Pr. + MR.	13.38 ± 0.33	80.53 ± 2.16	92.69 ± 2.00
Pr., T., MR.	**12.92± 0.42**	**81.09 ± 2.96**	**101.2 ± 3.32**

3) Evaluation of the AdaBoost model

The model: Comparing Adaptive Boosting, also known as AdaBoost, with other models described above is valuable for benchmarking and understanding which models perform best under different conditions. This comparison helps in identifying the most resource-efficient model that balances operational costs with accuracy. Adaboost is particularly notable for its adaptability, as it focuses on previously misclassified instances and combines multiple weak learners to improve accuracy and robustness against overfitting. It is a collection of numerous decision trees, each serving as a weak learner and is slightly better than random guessing. The key feature of the AdaBoost algorithm lies in its adaptability, as it carries the gradient of preceding trees to subsequent ones, thereby refining the errors of the preceding tree. This iterative learning process, which occurs at each step, culminates in the creation of a robust learner. The final prediction is derived from the weighted average of predictions provided by each individual tree [43]. Furthermore, the algorithm operates in a manner where the insights gained by earlier trees are transmitted to subsequent trees. This strategic information transfer ensures that future trees focus their learning efforts on challenging-to-predict training samples [44].

Application & Results:

In this comparative method, we trained our model using data such as employee presence, weather, vacation periods, public holidays, as well as information on transport strikes and meeting room reservations. We then used the model to anticipate attendances over an 18-day period, from 06/09/2023 to 29/09/2023.

Using this model, we analyzed its errors by calculating the difference between the predicted value and the actual value for each day. We then normalized this difference by dividing it by the real value, enabling us to assess the relative gap between prediction and reality. We present the results in Table 4, where the AdaBoost gap, expressed as a percentage, signifies the difference between the real and predicted values of the AdaBoost model. Calculated by determining the percentage of this difference relative to the actual value and then multiplying by 100.

A further evaluation of the model is carried out, where, following the example of SARIMAX and LSTM, we present a table of errors according to the different choices of variables. The results are detailed in Table 5.

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TABLE 4. Discrepancy between real data and AdaBoost predictions results.

Name of The Day	Date	AdaBoost Gap	%
Wednesday	06/09/2023	-66	10%
Thursday	07/09/2023	-24	4%
Friday	08/09/2023	-58	15%
Monday	11/09/2023	34	5%
Tuesday	12/09/2023	25	3%
Wednesday	13/09/2023	108	15%
Thursday	14/09/2023	90	12%
Friday	15/09/2023	-31	7%
Monday	18/09/2023	148	22%
Tuesday	19/09/2023	202	26%
Wednesday	20/09/2023	145	20%
Thursday	21/09/2023	80	12%
Friday	22/09/2023	67	12%
Monday	25/09/2023	27	4%
Tuesday	26/09/2023	56	8%
Wednesday	27/09/2023	-59	9%
Thursday	28/09/2023	-29	4%
Friday	29/09/2023	-54	13%
Absolute mean		**72**	**11%**

TABLE 5. Table of AdaBoost model errors for each variable choice for prediction of four weeks. Pr.: for employee presence variable, T.: for temperature variable, P.: for precipitation and MR.: for meeting room reservations.

Choice of variables	MAPE	MAE	RMSE
Pr., T. + P. + MR.	9.15 ± 0.00	56.231 ± 0.00	63.698 ± 0.00
Pr., T. + P.	21.76 ± 0.00	137.302 ± 0.00	151.366 ± 0.00
Pr., T.	23.50 ± 0.00	149.134 ± 0.00	161.52 ± 0.00
Pr. + MR.	8.37 ± 0.00	49.52 ± 0.00	57.57 ± 0.00
Pr., T., MR.	9.17 ± 0.00	56.27 ± 0.00	63.38 ± 0.00

4) Results of functional models

Results of the application of the *ftsa* approach:

To account for the uncertainty of this model, we use the lower and upper bounds returned by the prediction function (via the *ftsa* package). Specifically, for each iteration, we calculate the difference between the actual value and these two prediction bounds, producing two sets of errors: one associated with the "lower bound" predictions, the other with the "upper bound" predictions. Next, we aggregate these values and calculate the mean of the errors corresponding to the lower and upper bounds, as well as their difference (the interval "length"). The confidence interval is then obtained by taking the average of all the errors (based on the lower and upper bounds), and representing this average ± half the total length of the interval. This approach makes it possible to quantify the estimated uncertainty around each metric (MAPE, MAE and RMSE), based directly on the prediction intervals generated by the forecast methods.

We applied the model described in Subsection III-A3 to the same application study, which covers 42 weeks of training data. Subsequently, we evaluated the predictions on the 4 last weeks of data, based on three error metrics: Mean Absolute Percentage Error (MAPE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). The methods evaluated include ETS (Exponential Smoothing State Space Model), ARIMA (AutoRegressive Integrated Moving Average), RWDrift (Random Walk with Drift), RW (Random Walk), and ARFIMA (AutoRegressive Fractionally Integrated Moving Average). The results are summarized in Table 6.

TABLE 6. Comparison table of errors for each evaluated method of ftsa for prediction of the last four weeks of data.

Method	MAPE	RMSE	MAE
ets	22.97 ± 8.35	146.33 ± 55.25	139.81 ± 56.03
arima	**22.36 ± 5.19**	**141.27 ± 36.70**	**135.63 ± 37.02**
rwdrift	23.24 ± 3.67	146.95 ± 26.26	141.17 ± 26.11
rw	23.26 ± 3.95	147.03 ± 28.07	141.27 ± 27.97
arfima	22.36 ± 5.19	141.27 ± 36.70	135.63 ± 37.02

Looking at the different methods, we can see that ARIMA and ARFIMA perform similarly, but slightly better than ets. The rw and rwdrift methods, which are simple reference methods, show inferior performance to the other methods. The ARIMA approach appears to be the best performing of the evaluated methods, with a MAPE of **22.36 ± 5.19**, an RMSE of **141.27 ± 36.70** and a MAE of **135.63 ± 37.02**.

Results of the application of the deep neural network: To compare the results with the deep learning model described in Subsection III-B, we apply it to our case study, testing it on the last four weeks of our database with various input variable selections. We tuned the hyperparameters of the model, including the number of layers, the feedforward network dimension of the transformer module, and the alpha (L2 regularization loss for the acceleration parameter), by calculating the error for each combination of hyperparameters. For the number of layers and the dimension of the feedforward network of the transformer module, we tested the values [1, 2, 4, 8, 16, 32, 64, 128, 256, 2048], and for alpha [0, 0.05, 0.1, 0.3, 0.5, 0.9, 1, 1.5, 10, 100]. Then, we evaluated the prediction results using the parameters that yielded the smallest error. In our case, the number of layers is set to 4, the feedforward network dimension is 2, and the alpha is 0.5. We present in Table 7 the different MAPEs for each variable selection choice.

Results of the application of RobFTS model: The functional data models presented in Section III are evaluated in this section. With the same procedure of evaluation, we assess the four models by comparing the errors of each individual model as well as the error of the mean predictions across all four models. In our case study, we evaluated our models by varying parameters and exogenous variables, such as weather data, meeting room reservations, and lagged data of people present in the building of our case study. Table 8 presents the results of the various models with the different variables, evaluating their MAPE, the RMSE, and the MAE.

From Table 8, we observe that the combination *"Order(1), T., and MR."* consistently yields the lowest error metrics across the majority of models. For instance, for the MCD model, this combination achieves **MAPE = 4.99 ± 0.01**, **RMSE = 36.39 ± 0.11**, and **MAE = 28.76 ± 0.09**. In contrast, other variable groupings generally produce higher errors. For example, when comparing "Order(1) and T." or "Order(1) and

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TABLE 7. Table of DFMM model errors for each variable choice for prediction of four weeks. T: for temperature variable, Order(l): for lagged data of order l of the number of people presented, P: for precipitation and MR. for meeting room reservations.

Choice of variables	MAPE	RMSE	MAE
Order(1), Order(2) and T.	10.39 ± 0.45	73.92 ± 3.34	67.82 ± 2.77
Order(1), T. and MR.	10.43 ± 0.76	74.02 ± 5.28	68.31 ± 5.19
Order(1) and T.	10.47 ± 0.62	73.38 ± 2.74	68.05 ± 3.53
Order(1) and MR.	**10.22 ± 0.73**	**72.73 ± 3.51**	**66.74 ± 4.02**
Order(2) and T.	11.37 ± 4.00	80.04 ± 23.78	73.62 ± 23.72
Order(1), T., P., and MR.	10.74 ± 0.29	74.46 ± 1.12	69.55 ± 1.72

TABLE 8. Table of several models and errors by choice of variables for prediction of four weeks. T: for temperature variable, Order(l): for lagged data of order l of number of people presented, P: for precipitation and MR. for meeting room reservations.

Choice of variables	Model	MAPE	RMSE	MAE
Order(1), T. and MR.	**MCD**	**4.99 ± 0.01**	**36.39 ± 0.11**	**28.76 ± 0.09**
	MM	5.08 ± 0.00	38.98 ± 0.00	29.59 ± 0.00
	S	5.50 ± 0.04	41.88 ± 0.34	32.15 ± 0.19
	tau	5.22 ± 0.05	40.28 ± 0.33	30.76 ± 0.17
	Mean of the 4 models	5.03 ± 0.01	39.01 ± 0.15	29.45 ± 0.05
Order(1), T., P. and MR.	MCD	12.43 ± 0.34	93.19 ± 2.77	79.38 ± 2.29
	MM	5.91 ± 0.12	32.96 ± 0.54	42.21 ± 0.75
	S	5.82 ± 0.14	41.71 ± 0.86	33.02 ± 0.77
	tau	6.52 ± 0.35	45.3 ± 1.92	37.21 ± 1.87
	Mean of the 4 models	6.58 ± 0.13	45.78 ± 0.83	37.93 ± 0.77
Order(1) and T.	MCD	13.69 ± 0.00	90.045 ± 0.00	84.082 ± 0.00
	MM	14.12 ± 0.00	91.76 ± 0.00	86.95 ± 0.00
	S	13.91 ± 0.00	91.45 ± 0.00	85.18 ± 0.00
	tau	13.57 ± 0.00	88.94 ± 0.00	83.48 ± 0.00
	Mean of the 4 models	13.83 ± 0.00	90.32 ± 0.00	84.93 ± 0.00
Order(1), Order(2) and T.	MCD	19.54 ± 0.03	131.63 ± 0.14	118.12 ± 0.17
	MM	16.29 ± 0.00	105.79 ± 0.00	100.39 ± 0.00
	S	16.058 ± 0.00	104.03 ± 0.00	97.68 ± 0.00
	tau	15.94 ± 0.00	103.69 ± 0.00	97.51 ± 0.00
	Mean of the 4 models	16.96 ± 0.01	110.25 ± 0.04	103.42 ± 0.04
Order(1) and MR.	MCD	5.18 ± 0.02	37.186 ± 0.26	29.96 ± 0.17
	MM	5.06 ± 0.00	39.31 ± 0.00	30.14 ± 0.00
	S	5.88 ± 0.00	42.92 ± 0.00	35.40 ± 0.00
	tau	5.47 ± 0.01	40.49 ± 0.05	32.89 ± 0.07
	Mean of the 4 models	5.23 ± 0.01	39.4 ± 0.08	31.29 ± 0.08
Order(2) and T.	MCD	15.83 ± 0.00	105.25 ± 0.00	99.42 ± 0.00
	MM	13.56 ± 0.00	91.51 ± 0.00	86.75 ± 0.00
	S	15.26 ± 0.00	101.17 ± 0.00	96.40 ± 0.00
	tau	14.58 ± 0.00	96.96 ± 0.00	92.31 ± 0.00
	Mean of the 4 models	14.81 ± 0.00	98.49 ± 0.00	93.72 ± 0.00

MR." for the MCD model, the respective MAPE values rise to 13.69 ± 0.00 (with RMSE = 90.05 ± 0.00, MAE = 84.08 ± 0.00) and 5.18 ± 0.02 (with RMSE = 37.19 ± 0.26, MAE = 29.96 ± 0.17). These findings confirm that including the *MR* variable enhances predictive performance. More specifically, adding MR to temperature and lagged data (Order(1)) notably lowers the forecast errors compared to using only T alongside lagged data. Conversely, incorporating precipitation (P) does not guarantee improvement. For instance, when *P* is included in the combination "Order(1), T., P., and MR." for the MCD model, the MAPE increases to 12.43 ± 0.34 substantially higher than 4.99 ± 0.01 for "Order(1), T., and MR." alone. In summary, integrating *meeting room reservations* (MR.) alongside *temperature* (T) and lagged data of *order 1* (Order(1)) consistently offers superior model performance compared to other variable combinations.

In terms of comparison, we compare the results of this model (RobFTS) with the AdaBoost model evaluated. The prediction results from Table 9 exhibit variability between different dates, suggesting nuanced behaviors in their predictive capabilities. For example, on Fridays, RobFTS tends to display a larger gap compared to AdaBoost, indicating potential challenges in accurately predicting patterns for end-of-week scenarios. In contrast, on Mondays, AdaBoost generally exhibits a significant discrepancy. For instance; on 18/09/2023, there is a gap of 148 person predicted, suggesting its sensitivity in capturing underlying trends at the beginning of the week. Analyzing the discrepancies by considering the day of the week reveals valuable insights into the strengths and weaknesses of each model, which can guide further refinements in predictive algorithms and enhance overall forecasting accuracy.

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TABLE 9. Discrepancy between real data, AdaBoost and RobFTS model predictions results.

Name of The Day	Date	AdaBoost Gap	%	RobFTS Gap	%
Wednesday	06/09/2023	-66	10%	-36	6%
Thursday	07/09/2023	-24	4%	12	2%
Friday	08/09/2023	-58	15%	-55	14%
Monday	11/09/2023	34	5%	8	1%
Tuesday	12/09/2023	25	3%	-16	2%
Wednesday	13/09/2023	108	15%	60	8%
Thursday	14/09/2023	90	12%	67	9%
Friday	15/09/2023	-31	7%	-1	0%
Monday	18/09/2023	148	22%	8	1%
Tuesday	19/09/2023	202	26%	28	4%
Wednesday	20/09/2023	145	20%	10	1%
Thursday	21/09/2023	80	12%	-8	1%
Friday	22/09/2023	67	12%	69	12%
Monday	25/09/2023	27	4%	-1	0%
Tuesday	26/09/2023	56	8%	1	0%
Wednesday	27/09/2023	-59	9%	-66	11%
Thursday	28/09/2023	-29	4%	-14	2%
Friday	29/09/2023	-54	13%	-63	16%
Absolute mean		**72**	**11%**	**28**	**5%**
N° of days > 5%		**13**		**7**	

E. PREDICTION OF LAST WEEK OF DATA

In terms of comparison, we evaluated the different approaches for predicting the last week of data.

1) Comparison with the CatBoost model

In [21], the CatBoost model is applied as outlined in the introduction section. We have also applied this model to our case study to compare its results and performance. We trained the model using employee presence, weather and meeting oom reservations. We evaluated the results, based on the three error metrics: MAPE, MAE and RMSE. We present the results in Table 10.

TABLE 10. Comparison table of errors for the CatBoost model based on variable selection for prediction of last week of data. Pr.: for employee presence variable, T.: for temperature, P.: for precipitation and MR.: for meeting room reservations.

Variables	MAPE	RMSE	MAE
Pr.	26.60 ± 0.18	197.15 ± 129.69	155.37 ± 103.65
Pr., T.	26.78 ± 0.17	194.33 ± 124.34	155.87 ± 97.82
Pr., MR.	19.61 ± 0.11	133.19 ± 76.40	110.53 ± 63.17
Pr., T. + and P.	16.91 ± 0.07	107.53 ± 41.07	92.01 ± 37.33
Pr., T., and MR.	17.60 ± 0.09	115.77 ± 54.43	98.64 ± 48.57
Pr., T., MR. and P.	**13.14 ± 0.01**	**85.87 ± 7.38**	**67.22 ± 6.95**

2) ftsa predictions for one last week

TABLE 11. Table of several methods of ftsa model and errors for prediction of last week.

Method	MAPE	RMSE	MAE
ets	23.45 ± 5.53	140.92 ± 26.28	137.04 ± 29.87
arima	**23.03 ± 2.55**	**138.54 ± 8.37**	**134.33 ± 11.48**
rwdrift	23.85 ± 8.37	143.46 ± 43.52	139.44 ± 47.38
rw	23.87 ± 7.90	143.56 ± 40.67	139.59 ± 44.47
arfima	23.03 ± 2.55	138.54 ± 8.36	134.33 ± 11.47

In Table 11, both the ARIMA and ARFIMA methods stand out with the lowest error metrics. They share similar performance levels, each attaining a MAPE of **23.03 ± 2.55**, an RMSE of about **138.54 ± 8.36/8.37**, and an MAE of **134.33 ± 11.47/11.48**. These values indicate higher predictive accuracy relative to other methods. The ets approach shows moderate performance, with a MAPE of 23.45 ± 5.53, an RMSE of 140.92 ± 26.28, and an MAE of 137.04 ± 29.87. In contrast, the RWDrift and RW methods yield the highest error values, exceeding those of ARIMA, ARFIMA, and ets. Specifically, RWDrift obtains a MAPE of 23.85 ± 8.37, an RMSE of 143.46 ± 43.52, and an MAE of 139.44 ± 47.38, while RW reports similar metrics, indicating the lowest predictive accuracy.

3) DFMM predictions of the last week

Table 12 compares various combinations of variables in DFMM model for predicting the data for the last week. The tested variables are temperature (T.), first- and second-order lagged data (Order(1) and Order(2)), precipitation (P.), and meeting room reservations (MR.). Three error metrics–MAPE, RMSE, and MAE–evaluate the performance of each combination.

The most effective combination is "Order(1), T., P., and MR.", which achieves the lowest error values overall: a MAPE of **6.50 ± 1.03**, an RMSE of **48.10 ± 10.50**, and an MAE of **41.66 ± 7.73**. By comparison, "Order(1) and MR." also performs well, with a MAPE of 6.58 ± 0.69, an RMSE of 49.37 ± 2.59, and an MAE of 42.35 ± 3.51, confirming that meeting room reservations can substantially boost predictive accuracy. In conclusion, "Order(1), T., P., and MR." appears as the strongest predictor, underlining the significance of integrating precipitation, meeting room reservations, and first-order lagged data with temperature for more accurate DFMM forecasts.

4) RobFTS predictions of the last week

Table 13 compares various RobFTS methods (MCD, MM, S, and tau), each evaluated under different variable combinations. Across these scenarios, “Order(1), T., P., and MR.” stands out for its strong overall performance, especially when considering the mean of the four methods (MAPE = **4.56 ± 0.05**, RMSE = **33.18 ± 0.40**, and MAE = **23.39 ± 0.30**). This indicates that integrating precipitation (P.° in addition to temperature (T.), meeting room reservations (MR.) and first-order lagged data (Order(1)) contributes significantly to improving prediction accuracy. The combination of “Order(1), T., and MR.” demonstrates promising results, with the tau model achieving a notably low MAPE of 4.63 ± 0.01. In comparison, the configuration “Order(1) and MR.” yields robust performance for the MCD and MM models, with a MAPE of approximately 4.69. Conversely, the combinations “Order(1) and T.” and “Order(1), Order(2), and T.” exhibit higher average errors, suggesting that incorporating second-order lagged data (Order(2)) or relying exclusively on temperature (T.) without MR. or precipitation does not

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TABLE 12. Table of DFMM model errors for each variable choice for prediction of last week. T: for temperature variable, Order(l): for lagged data of order l of the number of people presented, P: for precipitation and MR. for meeting room reservations.

Choice of variables	MAPE	RMSE	MAE
Order(1), Order(2) and T.	7.52 ± 1.02	54.61 ± 7.03	47.90 ± 6.30
Order(1), T. and MR.	7.20 ± 1.06	51.32 ± 7.04	45.47 ± 6.73
Order(1) and T.	7.51 ± 1.04	55.97 ± 7.21	48.46 ± 6.44
Order(1) and MR.	6.58 ± 0.69	49.37 ± 2.59	42.35 ± 3.51
Order(2) and T.	7.60 ± 1.75	53.54 ± 9.48	48.15 ± 10.13
Order(1), T., P., and MR.	**6.50 ± 1.03**	**48.10 ± 10.50**	**41.66 ± 7.73**

TABLE 13. Table of several models and errors by choice of variables for prediction of last week. T: for temperature variable, Order(l): for lagged data of order l of number of people presented, P: for precipitation and MR. for meeting room reservations.

Choice of variables	Model	MAPE	RMSE	MAE
Order(1), T. and MR.	MCD	4.88 ± 0.04	35.163 ± 0.08	25.19 ± 0.25
	MM	4.76 ± 0.00	33.52 ± 0.00	24.77 ± 0.00
	S	4.93 ± 0.12	34.58 ± 0.95	25.66 ± 0.57
	tau	4.63 ± 0.01	35.01 ± 0.26	23.39 ± 0.14
	Mean of the 4 models	4.72 ± 0.03	34.44 ± 0.25	24.23 ± 0.25
Order(1), T., P. and MR.	MCD	4.73 ± 0.00	29.60 ± 0.00	25.85 ± 0.00
	MM	4.97 ± 0.00	35.33 ± 0.00	25.74 ± 0.01
	S	5.24 ± 0.01	36.47 ± 0.13	27.16 ± 0.02
	tau	5.15 ± 0.16	35.57 ± 1.04	27.16 ± 0.97
	Mean of the 4 models	**4.56 ± 0.05**	**33.18 ± 0.40**	**23.39 ± 0.30**
Order(1) and T.	MCD	9.47 ± 0.00	57.79 ± 0.00	51.34 ± 0.00
	MM	8.20 ± 0.00	50.24 ± 0.00	44.97 ± 0.00
	S	7.58 ± 0.00	49.88 ± 0.00	39.81 ± 0.00
	tau	7.46 ± 0.00	47.50 ± 0.00	39.85 ± 0.00
	Mean of the 4 models	8.18 ± 0.00	50.99 ± 0.00	43.99 ± 0.00
Order(1), Order(2) and T.	MCD	11.37 ± 0.00	66.98 ± 0.01	63.42 ± 0.01
	MM	8.20 ± 0.00	50.20 ± 0.00	45.02 ± 0.00
	S	7.39 ± 0.00	49.66 ± 0.00	38.52 ± 0.00
	tau	7.15 ± 0.00	47.77 ± 0.00	37.38 ± 0.00
	Mean of the 4 models	8.53 ± 0.00	52.64 ± 0.00	46.08 ± 0.00
Order(1) and MR.	MCD	4.69 ± 0.00	34.12 ± 0.00	24.04 ± 0.00
	MM	4.69 ± 0.00	33.62 ± 0.00	24.13 ± 0.00
	S	5.88 ± 0.00	39.07 ± 0.00	30.87 ± 0.00
	tau	5.83 ± 0.03	38.91 ± 0.13	30.54 ± 0.16
	Mean of the 4 models	5.23 ± 0.01	36.32 ± 0.03	27.11 ± 0.04
Order(2) and T.	MCD	6.21 ± 0.00	47.27 ± 0.00	40.28 ± 0.00
	MM	5.20 ± 0.00	32.37 ± 0.00	31.68 ± 0.00
	S	6.39 ± 0.00	46.27 ± 0.00	40.73 ± 0.00
	tau	6.10 ± 0.00	41.85 ± 0.00	38.31 ± 0.00
	Mean of the 4 models	5.98 ± 0.00	41.54 ± 0.00	37.75 ± 0.00

consistently enhance predictive accuracy. In summary, the results highlight the critical role of selecting appropriate variable combinations–particularly temperature, first-order lagged data, precipitation, and meeting room reservations–in enhancing the predictive accuracy of RobFTS model. Finally, this model appears to be the most robust and the one with the smallest deviation from the confidence intervals.

F. SELECTION OF THE MODELS

Following the presentation of the predictive results of the models, we now turn to the issue of model selection. While performance metrics such as RMSE or MAPE inform us about short-term forecasting accuracy, they do not necessarily account for model complexity. To complement this evaluation, we employ classical model selection criteria, including the Akaike Information Criterion (AIC) [45], the Bayesian Information Criterion (BIC) [46], the Hannan-Quinn Information Criterion (HQIC) [47], the Corrected Akaike Information Criterion (AICc) [48], and Sawa's Bayesian Information Criterion (SBIC) [49].

1) For statistical models (SARIMAX, RobFTS, ftsa)

These models are based on parametric formulations and allow for exact computation of the log-likelihood. Thus, the classical definitions of information criteria apply, using the number of estimated parameters K and the number of observations n:

$$\mathrm{AIC} = -2\ln\mathcal{L} + 2K \tag{7}$$

$$\mathrm{BIC} = -2\ln\mathcal{L} + K\ln(n) \tag{8}$$

$$\mathrm{AICc} = \mathrm{AIC} + \frac{2K(K+1)}{n-K-1} \tag{9}$$

$$\mathrm{HQIC} = -2\ln\mathcal{L} + 2K\ln(\ln(n)) \tag{10}$$

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Under the assumption of Gaussian residuals, the natural logarithm of the likelihood can be approximated by:

$$\ln \mathcal{L} \approx -\frac{n}{2}\left(\ln(2\pi) + \ln(\hat{\sigma}^2) + 1\right) \tag{11}$$

These formulations ensure statistically grounded comparisons between nested parametric models and are particularly suitable for time series models such as SARIMAX or functional models using maximum likelihood estimation.

2) For non-statistical models (e.g., LSTM, DFMM, Adaboost)

Unlike statistical models, deep learning architectures do not meet the fundamental assumptions required for the application of classical information criteria such as AIC or BIC [50]. Specifically, the following limitations hinder their direct applicability:

- **Absence of an explicit likelihood**: Deep models typically do not yield a tractable likelihood function.
- **Ambiguity in defining the number of parameters** (K): Regularization, dropout, and parameter sharing challenge the meaningful count of effective parameters.
- **Violation of asymptotic assumptions**: Many deep models operate in regimes where the number of parameters exceeds the number of training samples ($K \gg n$), invalidating asymptotic justifications of criteria such as AIC or BIC.
- **Computational burden**: Estimating criteria on multiple architectures or data splits is prohibitively expensive in practice.

Despite these incompatibilities, several empirical studies (e.g., [51, 52, 53]) propose practical adaptations. In particular, it is common to approximate the negative log-likelihood using the Mean Squared Error (MSE), under the assumption of Gaussian noise, and to substitute the number of trainable weights as a proxy for model complexity. This yields a heuristic—but still informative—framework for model comparison.

Under these approximations, the Equation (11) is reformulated as follows:

$$\ln \mathcal{L} \propto -\frac{|\mathcal{S}_{\text{test}}|}{2} \ln\left(\frac{\text{SSE}}{|\mathcal{S}_{\text{test}}|}\right), \tag{12}$$

where SSE denotes the sum of squared prediction errors:

$$\text{SSE} = \sum_{i=1}^{|\mathcal{S}_{\text{test}}|} \left(Y^{(i)} - \hat{Y}^{(i)}\right)^2. \tag{13}$$

Since SBIC was not presented previously, we provide its expression here as it was used to compare all models:

$$\text{SBIC} = |\mathcal{S}_{\text{test}}| \ln\left(\frac{\text{SSE}}{|\mathcal{S}_{\text{test}}|}\right) + 2(K+2)q + 2q^2. \tag{14}$$

Here, K is the number of trainable parameters in the model, $|\mathcal{S}_{\text{test}}|$ is the size of the test set, and:

$$q = |\mathcal{S}_{\text{test}}| \cdot \frac{\sigma^2}{\text{SSE}},$$

where σ is the standard deviation of prediction errors.

Full results of all model configurations and information criteria are presented inTable 14.

3) Discussion

The application of classical model selection criteria (AIC, BIC, AICc, HQIC, SBIC) remains a valuable complement to traditional forecasting metrics such as MAPE or RMSE. In examining Table 14, we observe clear trends that support the use of these criteria in comparing configurations both within and across model families. Within statistical models, RobFTS consistently outperforms SARIMAX across all variable configurations, achieving substantially lower information criterion scores. For example, the configuration "Pr., T., and MR." yields the lowest AIC (384.92) and also presents one of the best scores across other criteria such as HQIC (398.39) and SBIC (349.82), suggesting a balanced trade-off between model fit and complexity. SARIMAX, by contrast, displays AIC values that are an order of magnitude higher, indicating poor relative fit or excessive penalization due to fewer parameters failing to capture complexity.

When shifting to non-statistical models, the limitations of classical criteria become more apparent, particularly for deep architectures like LSTM and DFMM, where parameter counts ($K > 18,000$ for LSTM and >1.8 million for DFMM) inflate BIC and SBIC to uninformative levels (e.g., SBIC reaching into the hundreds of thousands). Nonetheless, AIC and AICc still provide intra-model insights. For example, among the LSTM configurations, Pr. and MR." achieves the lowest AIC (36501.42), which corresponds with better predictive performance observed empirically. AdaBoost, a comparatively more parsimonious ensemble model, offers a more interpretable application of these criteria. The configuration Pr. and MR." once again emerges favorably with the lowest AIC (6501.84) and one of the lowest HQIC scores (10415.87) in its category.

These consistent rankings across information criteria and forecasting accuracy suggest that, despite theoretical limitations, these criteria retain practical utility, especially for model selection within a given class. However, caution must be exercised when comparing absolute values between families: the gap between RobFTS and DFMM in AIC (e.g., 384 vs. over 3.7 million) underscores how model complexity distorts these metrics. Therefore, while absolute comparisons across model families are discouraged, intra-family evaluations remain both valid and informative.

V. COMPARISON WITH PUBLIC DATA

The availability of large-scale datasets related to workplace occupancy and employee flows remains limited due to confidentiality and data sensitivity concerns. Given these restrictions, we opted to evaluate our model on alternative datasets that still capture patterns of occupancy dynamics. One such example is parking occupancy data, which serves as an indirect yet insightful proxy for space utilization trends.

In this section, we applied the best-performing model, the **RobFTS** model, as evaluated in the results section, to a public dataset presented in the study [54]. The dataset used for this study concerns the occupancy rate of parking lots across 30 distinct parking facilities. It contains 35,717 entries

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TABLE 14. Information criteria for different models given identical variable configurations. T.: temperature, P.: precipitation, MR.: meeting room reservations, Pr.: number of people.

Choice of variables	Model	K	AIC	BIC	AICc	HQIC	SBIC
Statistical models							
Pr. and T.	**RobFTS**	**15**	**406.28**	**433.38**	**422.83**	**416.38**	**384**
	SARIMAX	6	2939.32	2960.13	2939.68	2947.71	2353.45
Pr., T., P., and MR.	**RobFTS**	**25**	**383.72**	**428.88**	**452.14**	**400.55**	**970.93**
	SARIMAX	9	2719.41	2750.62	2720.17	2731.99	2147.08
Pr., T., and MR.	**RobFTS**	**20**	**384.92**	**421.06**	**419.92**	**398.39**	**349.82**
	SARIMAX	8	2719.97	2747.72	2720.58	2731.15	2149.18
Pr. and MR.	**RobFTS**	**15**	**375.19**	**402.29**	**391.74**	**385.29**	**619.53**
	SARIMAX	7	2718.13	2742.41	2718.60	2727.92	2148.83
--	**ftsa**	**3**	**310.21**	**315.63**	**310.79**	**312.23**	**482.42**
Non-statistical models							
Pr. and T.	LSTM	18026	36500.08	69066.97	357.86	48640.69	389097.65
	DFMM	1845020	3690518.58	7023847.01	388.58	4933149.95	7023620.43
	AdaBoost	**1845**	**5130.73**	**876.05**	**11590.55**	**7732.09**	**133662.35**
Pr., T., P., and MR.	LSTM	18426	37302.28	70591.84	360.06	49712.29	410390.79
	DFMM	1870635	3741751.72	7121357.81	391.72	5001634.94	7121131.23
	AdaBoost	**2924**	**6861.36**	**478.75**	**17099.04**	**10984.06**	**269076.79**
Pr., T., and MR.	LSTM	18226	36901.78	69830.01	359.56	49177.09	408472.68
	DFMM	1870635	3741757.32	7121363.40	397.31	5001640.53	7121136.82
	AdaBoost	**2888**	**6753.22**	**442.00**	**16864.85**	**10825.16**	**271386.39**
Pr. and MR.	LSTM	18026	36501.42	69068.31	359.20	48642.03	405139.72
	DFMM	1845020	3690525.19	7023853.61	395.19	4933156.56	7023627.03
	AdaBoost	**2776**	**6501.84**	**412.62**	**16221.33**	**10415.87**	**270579.31**

collected between October 4 and December 19, 2016. Each record includes fields such as the parking system, capacity, occupancy, date, and time. The data was collected every 30 minutes, but this study has aggregated it by day.

The study uses several machine learning algorithms to predict available parking spaces, including **AdaBoost** and **TAdaBoost** (a modified version of AdaBoost that uses Tukey's biweight estimator to improve robustness). The performance of the algorithms is measured using several metrics: mean absolute error **MAE** and **RMSE**. Their TAdaBoost model achieved the following results on the metrics, with a MAE of **25.1** and an RMSE of **1811.6**.

We applied our **RobFTS** model to the same data to compare the results, and the outcomes are presented in Table 15. Based on the results presented in Table 15, we observe that the **RobFTS** model with the "tau" estimator outperforms the other models, including the **TAdaBoost** model presented in the study [54].

TABLE 15. Table of various errors from the application of the RobFTS model on the parking dataset.

Model	MAE	RMSE
MM	13.56 ± 0.00	14.22 ± 0.00
tau	$\mathbf{12.40 \pm 0.00}$	$\mathbf{13.10 \pm 0.00}$

VI. DISCUSSION

A. LIMITATIONS AND COMPUTATIONAL COST

1) Scalability of the RobFTS approach

We have examined the evolution of the MAPE as a function of the number of training weeks, setting a prediction horizon of one week. Table 16 presents the MAPE values corresponding to an increasing number of training weeks. We started our study from 30 weeks of training because, before this point, the model encountered a singularity issue in the system, indicating that we did not have enough data.

It can be seen that the error tends to decrease as the volume of training data increases. However, occasional error peaks appear at certain training durations, suggesting underlying phenomena such as seasonal changes, variations in employee behavior, or abnormal periods (e.g., strikes or vacations). In general, after 40 weeks of training, the model appears to achieve relative stability with a noticeable improvement in predictions, highlighting the need for a sufficient historical data record to guarantee reliable prediction.

TABLE 16. Evolution of MAPE as a function of the number of weeks of training.

N° of weeks of training	MAPE
30	16.323
31	8.668
32	4.034
33	19.094
34	6.446
35	4.446
36	4.276
37	10.995
38	23.622
39	14.353
40	6.672
41	3.778
42	6.423
43	3.622
44	3.874
45	4.416

Regarding the adaptability of our approach, we estimate that the RobFTS model can be applied to other real estate building databases. As long as the same types of input data are available, such as meeting room reservations and presence

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indicators derived from access control data (e.g., badge entry records), our approach can be transferred and implemented in new buildings. This flexibility enables the model to be generalized to different real estate environments.

2) Computational cost

The computational experiments for training and evaluating the RobFTS model were conducted on an HP ZBook Power 15.6 inch G10 Mobile Workstation, equipped with an Intel Core i7-13700H (Intel64 Family 6 Model 186 Stepping 2) running at 2.4 GHz and 32 GB of DDR4 RAM. The operating system used was Windows 11 Professional (Build 22631). Given the configuration of the workstation hardware, the training process required **1.41 seconds** to process 42 weeks of data, while inference on a single test instance (e.g. last 4 weeks) took approximately **0.02 seconds**. These metrics indicate that RobFTS can be efficiently trained on a high-end workstation without requiring specialized hardware such as GPUs, making it suitable for real-world deployment on enterprise-level systems.

B. REAL ESTATE INDUSTRY CONSTRAINTS

The application of artificial intelligence and predictive modeling in commercial real estate is still underdeveloped. Unlike other sectors such as finance or healthcare, the exploitation of real-time data and the use of advanced predictive analysis models in buildings are still poorly documented topics. This is partly due to the lack of data standardization and the heterogeneity of available data sources (presence sensors, room reservations, badge access data, weather conditions, etc.). The lack of homogeneous integration of this data prevents the rapid adoption of more effective predictive models. Furthermore, organizations exhibit a reluctance to share their occupancy data for reasons of confidentiality and compliance with General Data Protection Regulation (GDPR) requirements. This limitation restricts the development of large-scale datasets essential for refining and improving predictive models.

C. OPTIMIZING WORKSPACES: AN ECONOMIC CHALLENGE

A workstation in a company costs an average of 15.000 euros a year [15], but is occupied less than 50% of the time. This under-utilization is even more marked in commercial buildings, where the busiest days are mainly Tuesdays and Thursdays. This raises a major economic inefficiency, as workspaces remain unoccupied for long periods, while continuing to generate fixed costs. Using predictive models such as RobFTS, we intend to validate how predictive workplace presence modeling can optimize occupancy by anticipating high and low occupancy days.

D. PERSPECTIVES

Our results show that prediction accuracy is highly dependent on data volume and quality. A possible improvement would be to incorporate more exogenous variables, such as macroeconomic indicators, sectoral trends, or other transportation-related variables, to refine predictions.

In addition, the integration of adaptive learning to adjust models in line with changing work habits (telecommuting, new hybrid work modes) could reinforce the robustness of predictions. Another promising direction would be to train models on multiple regimes, such as periods with and without holidays, when enough data is available. This approach would help better capture variations in attendance patterns and mitigate the loss of training samples when filtering out days with less than 8% occupancy.

Finally, recent advances in Transformer-based architectures have demonstrated superior performance compared to LSTM models in time series forecasting tasks, particularly due to their ability to capture long-range temporal dependencies through self-attention mechanisms [55], [56], [57]. Integrating such models in future work could further improve accuracy and model generalization, especially in highly non-stationary environments. Notably, the DFMM model evaluated in this study is also based on a Transformer architecture, as illustrated in Figure 2, where self-attention is leveraged to model temporal evolution in a functional latent space. This highlights the relevance of Transformer mechanisms even in functional data contexts, and motivates their broader integration in hybrid deep learning frameworks.

VII. CONCLUSION

Based on the results presented in the results section, we conclude that the most effective model, which produces the best results and has the smallest error, is the RobFTS. Our results show that RobFTS outperforms standard ML and DL models in both MAPE and RMSE. Furthermore, the use of model selection criteria such as AIC and SBIC, even for non-parametric models, provides an interpretable and rigorous selection framework. The Fig.5 illustrates the comparison between the actual data and the best predictions generated by the various methods evaluated in this study, selecting models with the smallest error. We also present in Fig.6 the MAPEs of each model evaluated against each week for the prediction of the four last weeks. Based on the various models evaluated, it appears that the RobFTS model is the most effective at capturing variations in employee attendance. In contrast, for the DFMM model, it can be hypothesized that increasing the amount of data could enhance its performance and reduce its error rate in the test set. As for the ftsa model, its main limitation lies in its inability to integrate exogenous variables. This limitation is unfortunate because incorporating these variables could not only improve the performance of the model but also provide more comprehensive explanations by using additional factors that could enrich the model.

This study proposes to predict the time series of the number of employees in a specific workplace environment or in a building in general. An extension and generalization of the RobFTS model as outlined by [26], is proposed. This enhanced model incorporates the lagged variable of employee

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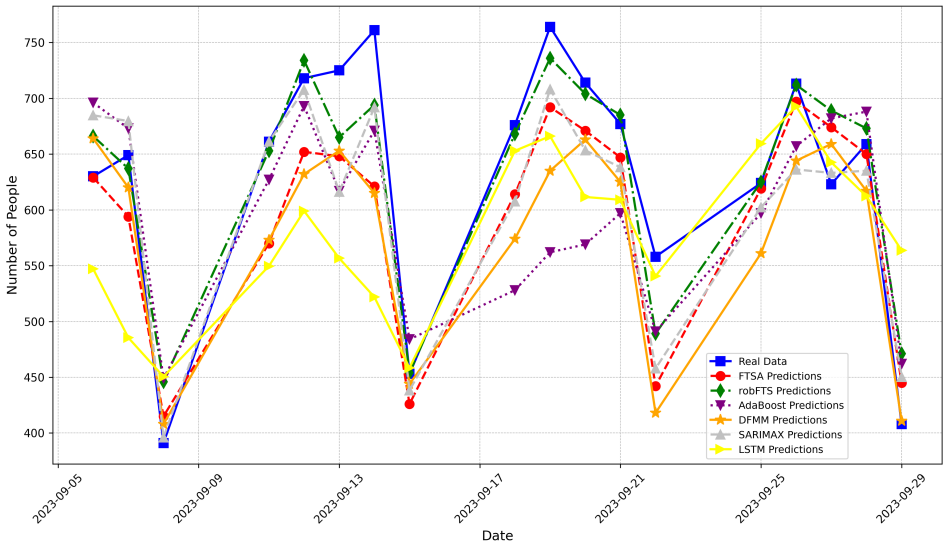


FIGURE 5. **The comparison of real data with the prediction of the models evaluated over the last four weeks of data.**

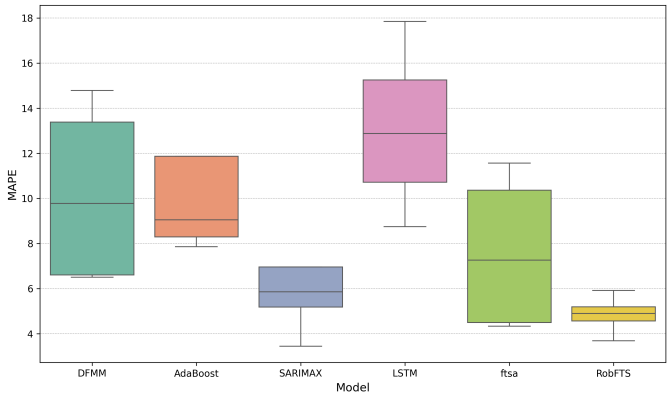


FIGURE 6. **MAPE variation over 4 weeks by model.**

presence and exogenous variables during the modeling phase. The results show that the presented model provides a better prediction accuracy than the other model.

Smart workplaces in general, and smart offices in particular enable data-driven decision making by providing real-time insights into various aspects of the workplace. By collecting and analyzing data on employee behavior and space utilization, businesses can identify areas for improvement and make informed decisions to optimize their operations. The concept of predictive modeling of real estate presence is still in its early stages, with few real-life case studies demonstrating significant cost reductions. However, interest in using data analytics to optimize office space utilization has been growing steadily in the real estate sector. Similar applications in the aviation and hospitality sectors - where predictive models forecast passenger numbers and hotel bookings - have proved highly effective in improving operational efficiency and minimizing wasted resources [58], [59]. These examples show that, with solid data collection and modeling, workplace presence prediction could also help companies objectify building use, optimize space allocation and, potentially, reduce real estate expenses. Looking forward, our research aims to expand on this potential by incorporating additional external factors, such as road traffic data including transportation conditions that can significantly impact the mobility of people and, consequently, their presence in workplaces. Through further case studies and simulations, we intend to validate how predictive workplace presence modeling can contribute to tangible cost reductions, resource efficiency, and sustainable energy management. In the long term, integrating these predictive technologies into smart workplaces presents a promising opportunity to meet the increase demand of the real estate industry for data-driven, cost-effective solutions.

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